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# Effect of nano-bubble water on high solid anaerobic digestion of pig manure: Focus on digestion stability, methanogenesis performance and related mechanisms

Yujie Fan[a], Zhongfang Lei[b], Xiaojing Yang[a], Motoyoshi Kobayashi[b], Yasuhisa Adachi[b], Zhenya Zhang[b], Kazuya Shimizu[b,*]

[a] Graduate School of Life and Environmental Sciences, University of Tsukuba, 1-1-1 Tennodai, Tsukuba, Ibaraki 305-8572, Japan
[b] Faculty of Life and Environmental Sciences, University of Tsukuba, 1-1-1 Tennodai, Tsukuba, Ibaraki 305-8572, Japan

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ABSTRACT

In this study, high solid anaerobic digestion of pig manure (PM) under nano-bubble water (NBW) addition was investigated with focus on digestion stability, methanogenesis performance and related mechanisms. Volatile fatty acids (VFAs) inhibition occurred when total solids (TS) was about 8% without NBW addition, which was alleviated with improved digestion stability under NBW addition, facilitating the process of high solids anaerobic digestion (HSAD). The cumulative $CH_4$ yield, on the other hand, was 201–230 mL/g-VS in the NBW reactors at TS of 3–6%, about 20.3–25.0% higher than the control reactors. At the same time, with higher water mobility and zeta potential, NBW was found to promote the consumption of soluble proteins/carbohydrates during the above AD process.

## 1. Introduction

The amount of pig manure (PM) produced in Japan is approximately 21 million tons per year (MAFF, 2019). Animal manure is generally treated by using wet anaerobic digestion (AD) which has been applied in the processing of livestock manure due to its many advantages (Kothari et al., 2014). The advantages of wet AD as a conventional method to treat animal manure has been less susceptible to inhibition and shorter lag phase, especially. However, wet AD still has many shortcomings when dealing with livestock manure including water-wasting, inconvenient operation, low energy recovery, and low fermentation slurry concentration (Chen et al., 2015). Therefore, in terms of economy and operability, it is necessary to find a more cost-effective and easy-to-operate method.

In comparison to wet AD with total solids (TS) 2–6%, high solids anaerobic digestion (HSAD), which often refers to AD with TS content higher than 8%, could be applied with more biogas production and less heat consumption (Liao and Li, 2015). The economic benefits of HSAD include not only the $CH_4$ produced from the AD process but also the digestate with the potential for fertilizer. HSAD, as an outstanding AD technology, has been widely used to treat various waste, such as chicken manure (Sun et al., 2016), dairy manure (Saady and Massé, 2015), pig manure (Huang et al., 2016; Wu et al., 2017), sewage sludge (Liu et al., 2016). However, it has some shortcomings, such as being susceptible to inhibition and longer lag phase (Chen et al., 2014). In addition, the high nitrogen content in PM could accumulate a large amount of ammonia nitrogen leading to cause ammonia inhibition in the AD process. The volatile solids (VS) removal rate of HSAD is also lower than that of conventional wet AD. Moreover, as the TS content increased, the mass transfer efficiency of the fermentation in the digester decreased and the rheological behavior of the digestates will be strongly affected, which adversely affected the use of the substrates by the microorganism (Garcia-Bernet et al., 2011). The microbial activity was also significantly reduced relative to wet AD (Liu et al., 2016). Many methods can be used to improve HSAD performance such as thermal hydrolysis process (Xue et al., 2015), percolate recirculation technology (Rico et al., 2015), continuous plug-flow dry AD reactor technology (Chen et al. 2015), low-temperature hydrothermal pretreatment (Huang et al., 2017). However, due to chemicals pollution and energy consumption or complicated operation, most of these methods have not been applied in industry. Although co-digestion can effectively avoid inhibition in HSAD process, it needs to add additional substrates with high C/N ratio, which increases the complexity of the operation and reduces the PM treatment capacity.

Water plays a very important role in the AD process because water promotes substrate hydrolysis and transfers the hydrolysate and other

* Corresponding author.
*E-mail address:* shimizu.kazuya.fn@u.tsukuba.ac (K. Shimizu).

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intermediates to the sites where the microorganisms can be utilized (Kothari et al., 2014). Nano-bubble water (NBW) has a large amount of nanoscale particle bubbles. Since nanobubbles have characteristics of great pressure, large specific surface area and minimal diameter, NBW has properties that are not possessed by ordinary water, such as strong adsorption, high air solubility, long storage time, etc. (Liu et al., 2019). Due to its unique characteristics, NBW has important value in the fields of industry (Kim et al., 2020), agricultural water management (Zhou et al., 2019), medicine (Hayashi et al., 2019), environmental pollution control (Xiao et al., 2019) and so on. One recent work showed that methane production was increased by 29% from waste activated sludge in AD process under $N_2$-nanobubble water addition (Yang et al., 2019). This work indicates that adding NBW has a positive effect in AD process to recover methane, but so far, few reports have been found to apply NBW in HSAD. Whether NBW can effectively avoid the shortcomings such as inhibition in the HSAD process is still unknown.

The aim of this research was to investigate the effect of NBW on HSAD, and to reveal a new path to enhance methane production and digestion stability of HSAD rather than using conventional methods. In order to evaluate the digestion stability of HSAD, this study measured the pH value, total alkalinity (TA), total ammonia nitrogen (TAN) and volatile fatty acids (VFAs). In addition, soluble organics in HSAD were also determined to assess HSAD performance. This work was the first study to reveal the effects of adding NBW on digestion stability, methanogenesis performance and the related mechanisms in the HSAD process of PM.

## 2. Materials and methods

### 2.1. Batch anaerobic fermentation experiments

#### 2.1.1. Raw manure and seed AD sludge

In this study, fresh PM was sampled from the domestic pig farm in Tsukuba, Ibaraki, Japan. The manure sample was collected manually to remove large particles. Then, the raw PM was stored at 4 °C. In this research, digested sludge was collected from a wastewater treatment plant located in Ibaraki, Japan, and then cultivated with the raw PM for acclimation of inoculum. Table 1 shows the main characteristics of inoculum and PM based on five tests in parallel.

#### 2.1.2. Batch AD experiments

Batch AD tests were carried out using two kinds of glass bottles. The 500 mL Scott Duran bottles were used for preparation and acclimation of inoculum and the 250 mL ones were applied for anaerobic digestion.

**Table 1**
Characteristics of raw swine manure used in this study.

| Parameters | Unit | Raw swine manure | Inoculum |
|---|---|---|---|
| Total solids (TS) | % | 41.10 ± 0.51 | 2.53 ± 0.09 |
| Volatile solids (VS) | % | 35.67 ± 0.34 | 1.72 ± 0.07 |
| Total nitrogen (TN) | mg/g | 13.07 ± 0.76 | 1.96 ± 0.13 |
| Total ammonia nitrogen (TAN) | mg/g | 4.21 ± 0.46 | 0.65 ± 0.04 |
| Total organic nitrogen (TON) | mg/g | 8.68 ± 0.57 | 119.39 ± 5.26 |
| Total alkalinity (TA, as $CaCO_3$) | mg/g-TS | 1.21 ± 0.07 | ND |
| Soluble total organic carbon (STOC) | mg/g-VS | 41.25 ± 0.95 | ND |
| Soluble proteins | mg/g-VS | 5.11 ± 0.80 | ND |
| Soluble carbohydrates | mg/g-VS | 2.79 ± 0.16 | ND |
| C | %/TS | 37.42 ± 0.58 | 45.30 ± 0.65 |
| H | %/TS | 5.79 ± 0.14 | 5.14 ± 0.21 |
| N | %/TS | 3.68 ± 0.12 | 3.90 ± 0.17 |
| C/N | – | 10.17 | 11.62 |
| pH | – | 6.08 ± 0.06 | 7.42 ± 0.05 |

ND (No detection).
Data in the table are expressed as Means ± SD.

The digested sludge was cultivated with the raw PM first under mesophilic condition (35 ± 2 °C) for about one month. And then the digestate was used as inoculum for the subsequent AD process. Four kinds of experimental reactors were set up: R(3% $TS)_{Control}$ and R(3% $TS)_{NBW}$ were loaded with 11 g PM + 89 g inoculum; R(6% $TS)_{Control}$ and R(6% $TS)_{NBW}$ were loaded with 25 g PM + 75 g inoculum; R(8% $TS)_{Control}$ and R(8% $TS)_{NBW}$ were loaded with 36 g PM + 64 g inoculum; R(10% $TS)_{Control}$ and R(10% $TS)_{NBW}$ were loaded with 46 g PM + 54 g inoculum. At the beginning of the experiments, 100 mL NBW was added into $R_{NBW}$ while 100 mL deionized water (DW) was added into $R_{Control}$ (Table 2). Additionally, four bottles loaded with inoculum (89 g, 75 g, 64 g, and 54 g, respectively) only were also tested in this work corresponding to four sets of experimental reactors. Biogas/methane production used in this research was obtained by subtracting the amount produced by the inoculum. To avoid microenvironment changes and to simplify this AD processes that might be applied in practice, in this study, no pH adjustment was conducted. All reactors were sealed with aluminum caps and rubber stoppers. Before placed in a temperature-controlled incubator (35 ± 2 °C), all the reactors were flushed with nitrogen gas for 2 min. After all reactors no longer produce gas for 4 consecutive days, the reaction stops. The reactors of R(3% $TS)_{Control}$ and R(3% $TS)_{NBW}$ were carried out for 36 days, while that of R(6% $TS)_{Control}$ and R(6% $TS)_{NBW}$ were 58 days. The reactors of R(8% $TS)_{NBW}$ were carried out for 68 days. Although R(8% $TS)_{Control}$, R(10% $TS)_{Control}$ and R(10% $TS)_{NBW}$ no longer produce gas in late HSAD process, for comparison with R(8% $TS)_{NBW}$, R(8% $TS)_{Control}$ were also carried out for 68 days. R(10% $TS)_{Control}$ and R(10% $TS)_{NBW}$ were conducted for 58 days. Each experiment was performed in duplicate. Shake all 250 mL anaerobic digesters manually for 2 min every day at a fixed time.

### 2.2. Preparation of NBW

NBW was prepared by a nano-bubble generator (HACK FB11, JAPAN) according to the previous report (Wang et al., 2019). Introduced carbon dioxide gas into deionized water (DW) through this generator at 20 ± 2 °C for 20 min to get NBW used throughout the AD process in this research. During the preparation of NBW, the pressure was controlled at 0.25 MPa and the flow rates of DW and carbon dioxide gas were kept at 8 L/min and 0.5 L/h, respectively. After preparation, NBW was used throughout the subsequent AD or HSAD process.

### 2.3. Analytical methods

VS, TS, TAN and total nitrogen (TN) in the PM samples were measured based on the standard method (APHA, 2012). As little nitrite or nitrate was detected in raw SM, total organic nitrogen (TON) was calculated by subtracting TAN from TN. The biogas and methane yield, pH and C/N ratio were determined according to the methods described in the report (Fan et al., 2020). In this research, accumulative biogas yield (mL/g-VS) and accumulative $CH_4$ yield (mL/g-VS) were used to assess AD process. Soluble carbohydrates and proteins were determined by using the phenol–sulfuric method and Lowry method, respectively. The alkalinity, including partial alkalinity (PA), intermediate alkalinity (IA) and TA were tested according to the reported methods (Sun et al., 2016). PA and TA were determined by an endpoint of pH 5.75 and 4.3, respectively. IA was measured from pH 5.75 to 4.3. As for VFAs determination, a GC-FID (GC-8A, SHIMADZU CO., Japan) was used. Soluble total organic carbon (sTOC) was measured by using a TOC analyzer (TOC-VCSN, ASI-V autosampler, SHIMADZU CO., Japan).

VS reduction rate (%) was determined according to the following Eq. (1)

$$VS\ \text{reduction rate}\,(\%) = \frac{[VS]_{initial} - [VS]_{final}}{[VS]_{initial}} \times 100 \quad (1)$$

**Table 2**
Experimental conditions for $R_{Control}$ and $R_{NBW}$ in this study.

| | SM (g) | Inoculum (g) | NBW (g) | DW (g) | $TS_{ea}$ (%) | $VS_{ea}$ (%) |
|---|---|---|---|---|---|---|
| R (3% TS) $_{NBW}$ | 11 | 89 | 100 | 0 | 3.18 ± 0.16 | 2.54 ± 0.05 |
| R (3% TS) $_{Control}$ | 11 | 89 | 0 | 100 | 3.25 ± 0.09 | 2.66 ± 0.10 |
| R (6% TS) $_{NBW}$ | 25 | 75 | 100 | 0 | 6.11 ± 0.13 | 4.72 ± 0.06 |
| R (6% TS) $_{Control}$ | 25 | 75 | 0 | 100 | 6.04 ± 0.06 | 4.64 ± 0.09 |
| R (8% TS) $_{NBW}$ | 36 | 64 | 100 | 0 | 8.10 ± 0.08 | 6.38 ± 0.04 |
| R (8% TS) $_{Control}$ | 36 | 64 | 0 | 100 | 7.94 ± 0.22 | 6.33 ± 0.10 |
| R (10% TS) $_{NBW}$ | 46 | 54 | 100 | 0 | 10.21 ± 0.20 | 8.53 ± 0.18 |
| R (10% TS) $_{Control}$ | 46 | 54 | 0 | 100 | 10.11 ± 0.15 | 8.39 ± 0.12 |

$TS_{ea}$ (Experimental actual TS content).
$VS_{ea}$ (Experimental actual VS content).

Free ammonia nitrogen (FAN) was determined based on the following Eq. (2)

$$\frac{FAN}{TAN} = \left(1 + \frac{10^{-pH}}{10^{-\left(0.09018 + \frac{2729.92}{T(K)}\right)}}\right)^{-1} \quad (2)$$

whereas $T$(K) is the temperature in Kelvin.

### 2.4. Statistical analysis

For determination of NBW characterization and AD experiments, mean values were calculated by adopting Microsoft Office Excel 2019 according to three parallel tests. For statistical analysis, one-way analysis of variance (ANOVA) was used. The significant difference in this study was set at $p < 0.05$.

## 3. Results and discussion

### 3.1. AD process stability

The pH value is the result of a combination of solubility equilibrium, liquid–solid $CO_2$ balance and acid-alkali balance during the AD process (Chen et al., 2015). As shown in Fig. 1a, at the beginning of the AD process, the pH value of all reactors dropped rapidly due to the large amount of acid generation in the acidogenesis stages. The initial pH values were different because with the increase of TS, the content of PM increased with the content of acidic substances increased. In addition, the raw material PM had a low pH of 6.08 (Table 1), which indicated that the raw material contained a large amount of acidic substances. The pH value in R(8% TS)$_{Control}$, R(10% TS)$_{Control}$ and R(10% TS)$_{NBW}$ was maintained at around 6.3, indicating that the AD process of this three kinds of reactors might be inhibited because of VFAs accumulation (Fig. 1c). Other reactors, however, gradually increased after the initial decline and remained at around 7.4 because VFAs finally converted to methane (Fig. 1b, c and Fig. 3).

The VFAs concentration of R(3% TS)$_{NBW}$ and R(3% TS)$_{Control}$ decreased rapidly as the AD process progressed. When TS was about 3%, the acetic acid was rapidly consumed and maintained at a lower level in the later stage. Similarly, propionic and butyric acids were quickly consumed in two weeks. Therefore, VFAs concentration and methane production indicated acidogenesis and methanogenesis stages proceeded fairly well when TS content was about 3%. As TS content increased, the initial VFAs concentration increased (Fig. 1b, c). When TS content was about 6%, the initial VFAs concentration was about 5,500 mg/L. Compared with the R(6% TS)$_{Control}$, R(6% TS)$_{NBW}$ experienced a faster increase rate in the initial stage and a faster consumption rate in the later stage. As the TS content was further increased to approximately 8%, the VFAs concentration in R(8% TS)$_{Control}$ was accumulated and was not consumed in the later stage of HSAD. The R(8% TS)$_{NBW}$, however, was consumed in the late HSAD after the initial accumulation. Regardless of R(10% TS)$_{Control}$ or R(10% TS)$_{NBW}$, VFAs concentration exceeded 10,000 mg/L after the 4th day, and they were not consumed in the later stage of HSAD after the accumulation. The most important phenomenon was that the pH value of R(8% TS)$_{NBW}$ gradually increased after day 20. These results showed NBW addition promoted the consumption of VFAs to covert methane, thereby restoring the stability of the AD system.

VFAs are related to alkalinity, pH and the activity of methanogens. Therefore, they are important indicators for evaluating the performance of AD processes (Abouelenien et al., 2014). In this work, the addition of NBW played a role in promoting the consumption of VFAs leading to maintain the stability of the HSAD system, avoiding serious inhibition due to VFAs accumulation. This phenomenon led to promote the recovery of the pH value of the HSAD system for the stability. However, when the concentration of VFAs was greater than the threshold, adding NBW could not promote the fact that the HSAD process failed. This was because the accumulation of VFAs leads to a decrease in pH value, and when pH decreases, the toxicity of VFAs increases (Saady and Massé, 2015). Moreover, when pH value is low (5–7), VFAs are present in an undissociated form, which is toxic for microorganisms and can specifically affect methanogens (Liew et al., 2011). In this work, when the concentration of VFAs was greater than 10,000 mg/L, R(10% TS)$_{NBW}$ could not be recovered in the late stage of HSAD (Fig. 1c). As for the pH value of R(10% TS)$_{NBW,}$ it was about 6.3, which was in the range of 5–7 (Fig. 1a). In terms of the threshold that NBW could withstand the inhibition from VFAs, further researches are needed.

Changes in TA content in all reactors are shown in Fig. 1d. In the late stage of AD, the TA content of $R_{Control}$ was significantly lower than that of $R_{NBW}$ when TS content was about 8% (Fig. 1d). However, in other TS cases, there was no significant difference between $R_{Control}$ and $R_{NBW}$. This indicated that when the HSAD system was slightly inhibited (for example, R(8% TS)$_{NBW}$), the addition of NBW could increase the TA content, which was conducive to the restoration of the stability of the HSAD system. However, when the HSAD system was severely suppressed (such as R(10% TS)$_{NBW}$), adding NBW had no significant effect on TA content.

IA indicates the impact from VFAs while PA shows the available pH buffering from ammonia and bicarbonate. Therefore, the IA/PA ratio can be used to indicate the stability of the AD system. When IA/PA suddenly increases, it indicates the pH buffering ability of AD decreases (Sun et al., 2016). Fig. 2a shows the ratio between IA/PA under different TS content. As the AD process proceeded, the IA/PA ratio of R(3% TS)$_{NBW}$, R(3% TS)$_{Control}$, R(6% TS)$_{Control}$, R(6% TS)$_{NBW}$ and R(8% TS)$_{NBW}$ was gradually decreasing. This showed that the stability of the above five groups was better. Since the IA/PA ratio was in the range of 3 to 5.5 and was not decreased in R(8% TS)$_{Control}$, R(10% TS)$_{Control}$ and R(10% TS)$_{NBW}$, these three kinds of reactors had the poor buffering ability. The different results of the IA/PA ratio of R(8% TS)$_{NBW}$ and R(8% TS)$_{Control}$ indicated that NBW addition could improve the stability of the HSAD system.

VFA/TA ratio is more reliable in detecting the stability of the AD process than simply detecting the pH value (Chen et al., 2014). When

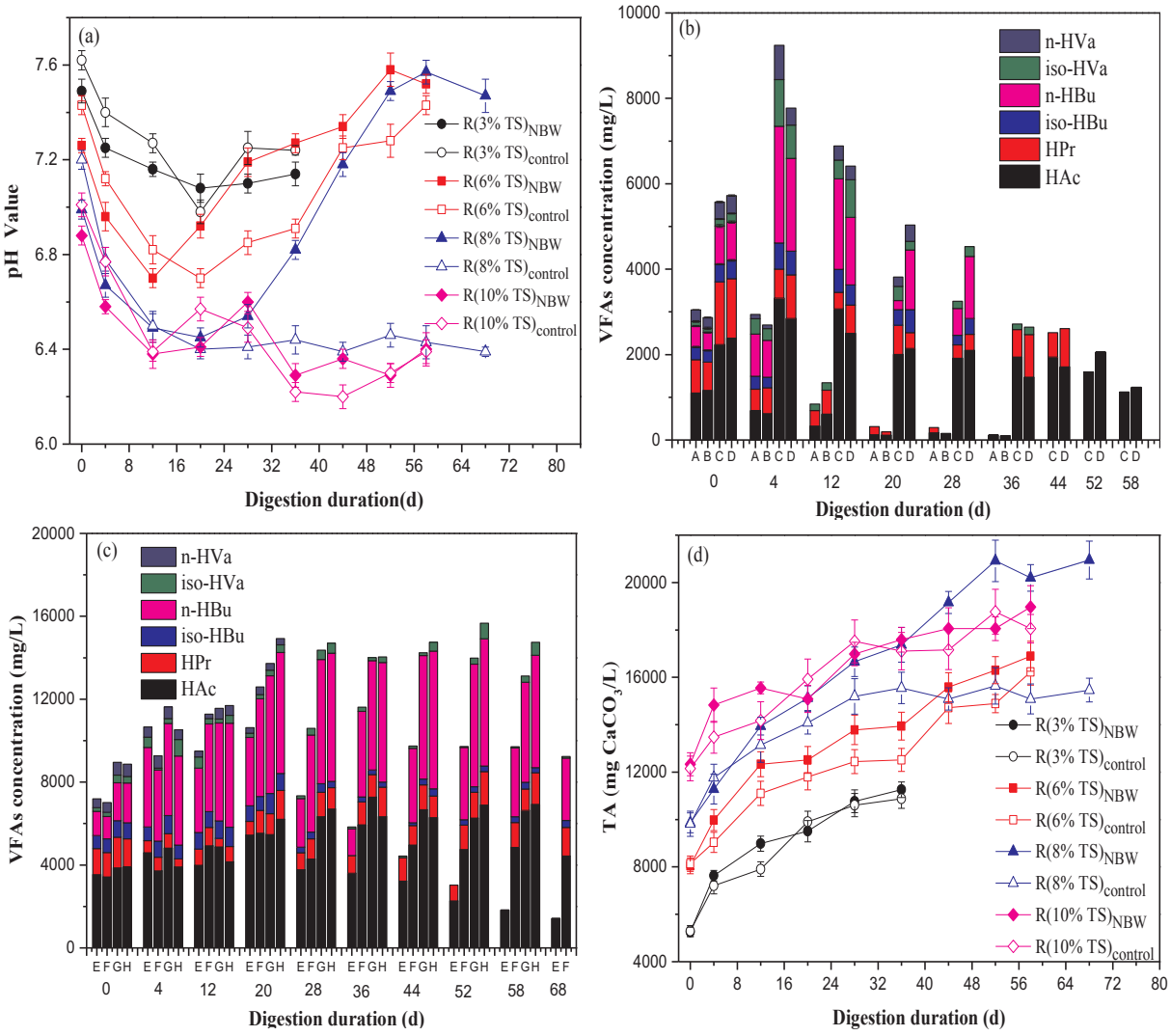



**Fig. 1.** Change of pH, VFAs concentration, and TA.

the ratio is lower than 0.4, the AD process is stable, which is the optimal for AD process. When the ratio is higher than 0.8, the digester is prone to significant instability (Li et al., 2013). Fig. 2b shows the VFA/TA ratio with or without NBW addition under different TS content. Since the VFA/TA ratio of R(8% TS)$_{NBW}$ was lower than 0.4 in the later stage of HSAD, this indicated that the HSAD process of R(8% TS)$_{NBW}$ gradually stabilized in the later stage. R(8% TS)$_{Control}$, however, its ratio was greater than 0.8, which indicated that R(8% TS)$_{Contro}$ did not return to stability in the late stage of the HSAD process. Therefore, VFA/TA ratio also showed that the addition of NBW improved the stability of HSAD.

Both TAN and FAN are important factors influencing methanogenic activity in the HSAD system. The TAN content of R(3% TS)$_{Control}$, R(3% TS)$_{NBW}$, R(6% TS)$_{Control}$, and R(6% TS)$_{NBW}$ did not exceed 1,500 mg/L throughout the AD process (Fig. 2c). That is, when TS was 3% and 6%, no ammonia inhibition occurred. When TS was about 8%, the TAN content gradually increased as the AD process progressed. The TAN content of R(8% TS)$_{NBW}$ was around 1,500 mg/L, while R(8% TS)$_{Control}$ was continuously increased to about 2,000 mg/L. Chen et al. (2008) reported that ammonia inhibition is possible when the TAN content exceeds 1,700 mg/L. Therefore, when TS was 8%, ammonia inhibition might occur in R(8% TS)$_{NBW}$ and R(8% TS)$_{Control}$. The TAN content also exceeded 1,500 mg/L when TS was about 10%, which indicated that ammonia inhibition might also occur in R(10% TS)$_{Control}$ and R(10% TS)$_{NBW}$. FAN is capable of penetrating the cell membrane of microorganisms and accumulating inside the cell to cause proton imbalance or toxic cytoplasmic enzymes. Therefore, FAN is more toxic than ionized ammonia (Niu et al., 2014). Except for R(6% TS)$_{Control}$, R(6% TS)$_{NBW}$ and R(8% TS)$_{NBW}$, the FAN concentration of other digesters did not exceed 50 mg/L throughout the AD process (Fig. 2d). Under the same conditions of TAN and temperature, a change in pH value of 0.1 will result in a 25% change in FAN concentration (Chen et al., 2015). Therefore, pH has a significant impact on FAN. Since R(8% TS)$_{Control}$, R(10% TS)$_{Control}$ and R(10% TS)$_{NBW}$ had the lower pH (around 6.3), the FAN content was also lower even though the TAN content was higher than other digesters. The inhibition threshold for FAN concentrations is reported to be 80–100 mg/L for conventional AD (De Baere et al., 1984). Therefore, the inhibition from FAN did not occur in all digesters.

### 3.2. Methanogenesis and soluble organics performance

Fig. 3 shows biogas and methane production of the entire AD process. When TS was 3% and 6%, the cumulative $CH_4$ and biogas yields from R$_{Control}$ were lower than those from R$_{NBW}$. When TS was 8%, the HSAD process of R$_{Control}$ was inhibited, and the biogas and $CH_4$ production were significantly reduced. In contrast, R$_{NBW}$ was relatively high with a TS of 8%, but its lag phase was relatively long compared to the lower TS series. When TS was 10%, the HSAD process of R$_{Control}$ and

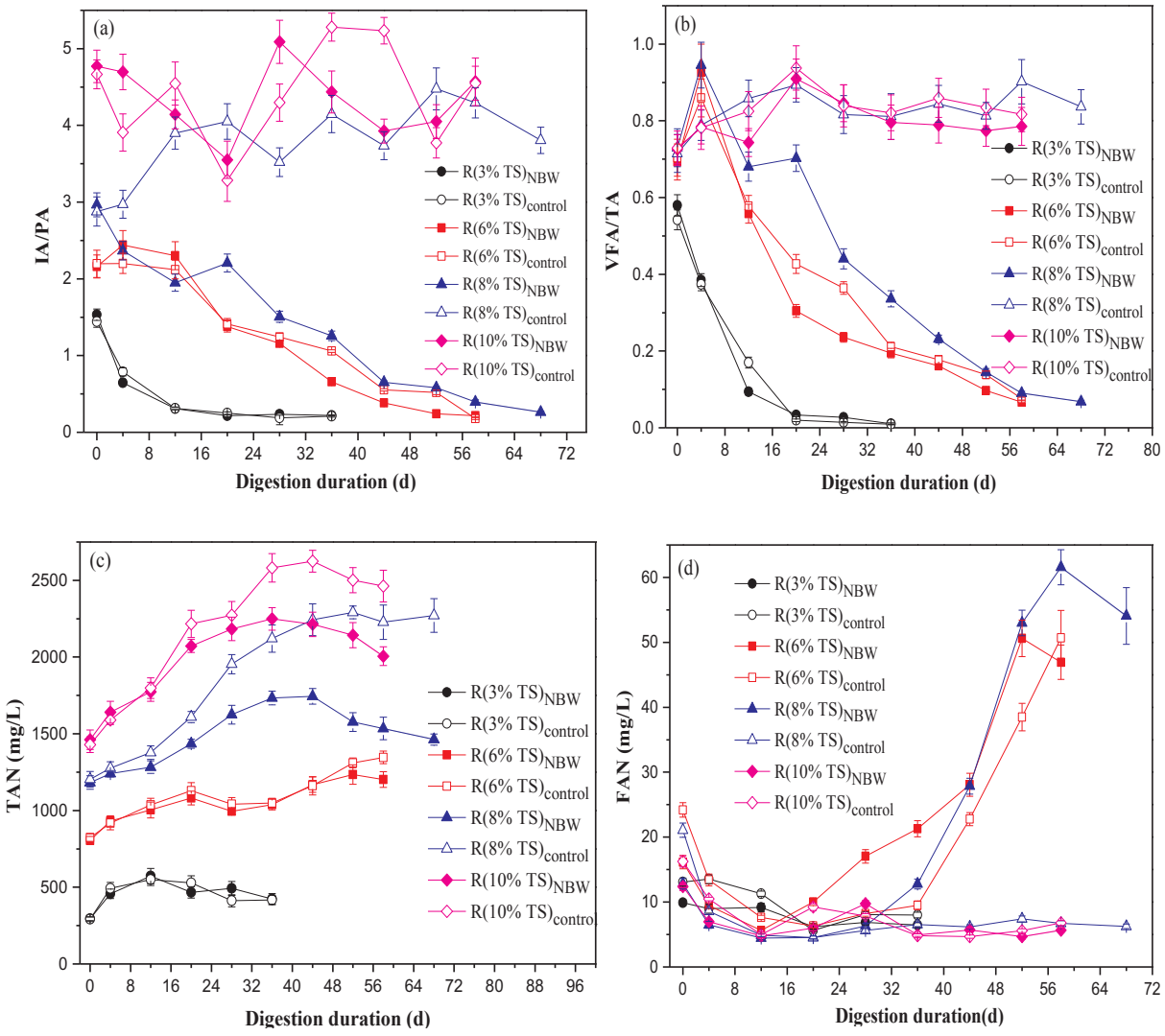



**Fig. 2.** Change of IA/PA, VFA/TA variation, and TAN variation.

$R_{NBW}$ was suppressed, and biogas and $CH_4$ yields were low. In terms of cumulative biogas production (Fig. 3a), R(3% TS)$_{NBW}$ and R(6% TS)$_{NBW}$ achieved 336 and 345 mL/g-VS, respectively, increasing by 17.5% and 23.9% in comparison to R(3% TS)$_{Control}$ and R(6% TS)$_{Control}$, respectively. R(8% TS)$_{NBW}$ obtained the highest biogas yield, which was 386 mL/g-VS. However, the corresponding R(8% TS)$_{Control}$ was only 62 mL/g-VS. This was due to that anaerobic bacteria, especially methanogens, were reported to be sensitive under acidic environment and their growth was inhibited in acidic conditions (Kothari et al., 2014). The pH value of R(8% TS)$_{Control}$ was low (around 6.3), and VFAs inhibition occurred during the HSAD process (Fig. 1). As a result, the activity of methanogens was affected, and biogas production was reduced. For the cumulative $CH_4$ production (Fig. 3b), R(3% TS)$_{NBW}$ and R(6% TS)$_{NBW}$ reached 201 and 230 mL/g-VS, respectively, corresponding to 20.3% and 25.0% increase when compared with R(3% TS)$_{Control}$ and R(6% TS)$_{Control}$, respectively. Similarly, R(8% TS)$_{NBW}$ obtained the highest $CH_4$ production, which was 261 mL/g-VS. R(8% TS)$_{Control}$, however, was only 23.8 mL/g-VS.

Previous reports showed the addition of NBW promote $CH_4$ production in AD process (Yang et al., 2019; Fan et al., 2020). In this study, although the $CH_4$ yield from $R_{NBW}$ was higher than $R_{Control}$, it was still lower than the theoretical methane potential from PM, which was 290 to 550 L-$CH_4$/kg-organic matter (Ferreira et al., 2014). The reason might be due to the presence of undegradable organics in PM, or it might be due to the difference in the inoculum. In addition, in this study, the methane increase rates of R(3% TS)$_{NBW}$ and R(6% TS)$_{NBW}$ were 20.3% and 25.0%, respectively, which was lower than the 34% methane increase rate achieved by low-temperature hydrothermal pretreatment reported by Huang et al (2017). Simultaneously, when TS content was greater than 10%, the effect of NBW technology was also inferior to the effect achieved by using a down plug-flow anaerobic reactor (Chen et al. 2015). When anaerobic digestion was used to treat animal manure with high TS, although the effect of NBW was inferior to low-temperature hydrothermal pretreatment technology and down plug-flow anaerobic reactor, NBW technology did not need to consume too much energy, nor did it need to use complex reactor. Moreover, in AD process, NBW technology also has advantages, especially low cost, low pollution, and simple operation.

In this work, when TS was < 8%, the addition of NBW increased $CH_4$ production. The results suggested NBW addition enhanced the hydrolysis process of organics such as soluble proteins/carbohydrates, which could promote the subsequent methanogenesis process and thus increase $CH_4$ yield. As the TS content increased to about 8%, the HSAD process might be inhibited. After a long lag phase, NBW addition can reduce the degree of inhibition, which has been proven in the changing trends of VFAs, pH, IA/PA ratio. When TS content was about 10%, the HSAD process was severely inhibited, and NBW addition could not change the HSAD process failed in this study. The reason may be that

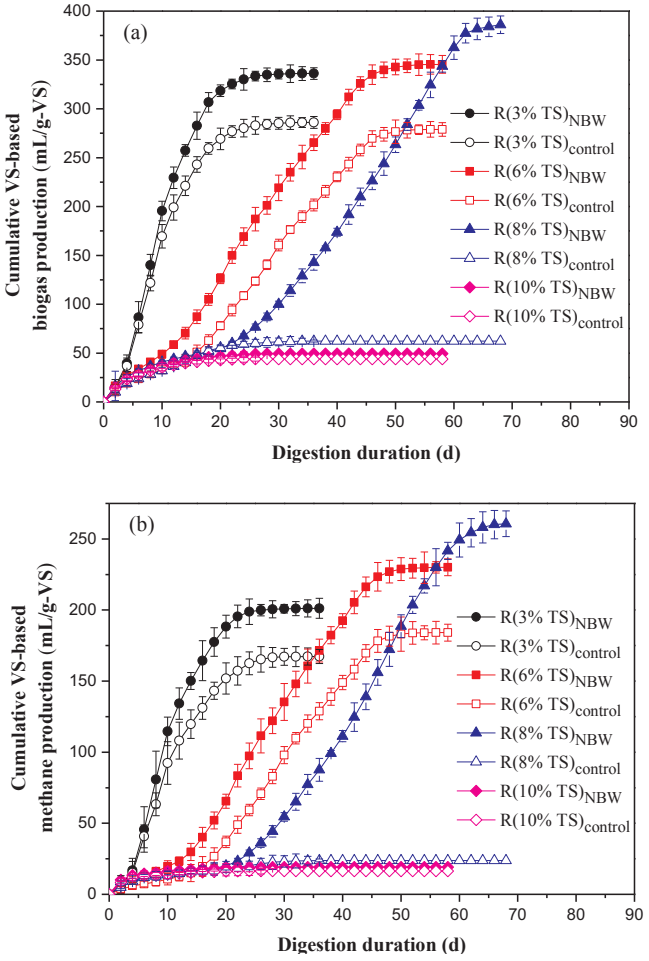



**Fig. 3.** Change of biogas production and methane production.

the accumulation of intermediate products, such as VFAs exceeds the threshold for NBW to withstand inhibition. In addition, when the TS was about 10%, The lower inoculum ratio and acidic PM used in this study may also be the reason for the high VFAs content and ultimately the failure of HSAD. Further analysis needs to be carried out to study the effect of NBW on methanogenesis of HSAD including gene expression dynamics.

Therefore, this work measured some soluble organics during AD process. As shown in Fig. 4a, the soluble proteins in $R_{NBW}$ increased rapidly and exceeded the corresponding $R_{Control}$ from the beginning to the day 4. As the AD progressed, when TS was about 3% and 6%, the soluble proteins were rapidly consumed in $R_{NBW}$ and $R_{Control}$, possibly due to the rapid reproduction of hydrolyzed bacteria during the early stages of AD (Yuan et al., 2017). The soluble proteins in R(8% TS)$_{NBW}$ dropped rapidly after day 20. However, R(8% TS)$_{Control}$ was not rapidly consumed, which indicated inhibition has occurred and caused the HSAD process to fail. When TS was about 10%, the soluble proteins of $R_{NBW}$ and $R_{Control}$ were not consumed in a large amount, which also showed that the HSAD process was not successful. In this work, the changing trends of soluble proteins and carbohydrates during AD process were basically similar (Fig. 4b). By analyzing soluble carbohydrates/proteins, it could be concluded that adding NBW could avoid the accumulation of soluble organics when there was no serious inhibition occurred in the HSAD process. However, when severe VFAs inhibition occurred in R(10% TS)$_{Control}$ and R(10% TS)$_{NBW}$, adding NBW could not effectively promote the consumption of soluble organics. When the TS was 3% and 6%, both $R_{NBW}$ and $R_{Control}$ decreased rapidly first and then slowly decrease in VS content (Fig. 4c). The difference is that the VS reduction rates in $R_{Control}$ on day 4 (9.8% in TS 3% and 3.6% in TS 6%) were slower than those in $R_{NBW}$ (15.9% in TS 3% and 6.4% in TS 6%). When TS was 8%, the VS reduction rates in $R_{NBW}$ was slower in the early stage of AD, but it became faster in the later stage. As for the R(8% TS)$_{Control}$, however, the VS reduction rates were slow during the AD process. Although the VS reduction rates of both R(10% TS)$_{NBW}$ and R(10% TS)$_{Control}$ were slow, R(10% TS)$_{NBW}$ were still faster than that of the R(10% TS)$_{Control}$ on day 4 (2.8% in $R_{NBW}$ and 1.9% in $R_{Control}$). However, with the further increase of VFAs content (Fig. 1c), the decrease of VS was no longer obvious when TS was about 10%. Therefore, when the inhibition was not serious, the addition of NBW accelerates the hydrolysis process, which has been reported (Yang et al., 2019). However, when the HSAD process of R(10% TS)$_{NBW}$ was severely inhibited (excessive VFAs inhibition), adding NBW could not significantly increase the rate of hydrolysis in this study. Fig. 4d illustrates the changes of sTOC throughout the AD process. Obviously, since there was no ammonia inhibition and VFAs inhibition when TS was about 3% and 6%, the sTOC was firstly increased and then decreased. When TS was 8%, in the early stage, the sTOC of R(8% TS)$_{NBW}$ and R(8% TS)$_{Control}$ accumulated continuously. With the HSAD progressed, the consumption rates of sTOC from R(8% TS)$_{NBW}$ was higher than the accumulation rates after a long lag phase, and the sTOC content gradually decreased. R(8% TS)$_{Control}$, however, was not rapidly consumed throughout the HSAD process. When TS was 10%, the sTOC content in both R(10% TS)$_{NBW}$ and R(10% TS)$_{Control}$ gradually stabilized after the initial increase. The changes of sTOC content further illustrated that the HSAD process of R(10% TS)$_{NBW}$ and R(10% TS)$_{Control}$ has been seriously inhibited. Meantime, when the inhibition was not serious in R(8% TS)$_{NBW}$ and R(8% TS)$_{Control}$, adding NBW could gradually alleviate the inhibition within a longer lag phase and promote the consumption of soluble organics.

### 3.3. Related mechanisms

To clarify the relevant mechanisms, the characteristics of NBW were analyzed in this work. Based on previous researches, $CO_2$-NBW have higher zeta potential, spin–spin relaxation times ($T_2$), particle density than DW (Wang et al., 2019, Wang et al., 2020). As known, zeta potential has an important effect on the stability of the bubble (Parmar and Majumder, 2013). When zeta potential is high (negative or positive), colloids tend to avoid coalescence and have the property of being electrically stable. Due to $CO_2$-NBW addition, zeta potential from $CO_2$-NBW was significantly higher than DW ($p < 0.05$). Therefore, $CO_2$-NBW addition may avoid the formation of aggregates in HSAD process. $T_2$ can be used to detect interactions between weak molecules such as hydrogen bonding, steric effects, and molecular mobility (Balci, 2005). Liu et al (2013) pointed out that the number of nano-bubbles has a positive correlation with the $T_2$ in water and the introduction of nano-bubbles can disturb the hydrogen-bonding network and make the water molecules have higher mobility. In this work, the $T_2$ value in $CO_2$-NBW was longer than that in DW, which indicated that the water molecules in the reactors of $R_{NBW}$ have higher mobility than that of $R_{Control}$. In addition, NBW with higher mobility could improve nutrient mass transfer efficiency from liquid to bacterial cells (Wang et al., 2019). This may be beneficial for improving the activity of microorganisms and enzymes.

Nano-bubbles are once formed, they are highly durable (Zimmerman et al., 2011). Wang et al. (2019) reported that the bubble particles were still stable after NBW storage for 14 days. Therefore, NBW can have an impact in HSAD in a rather long time, rather than only in a short period of time. When the inhibition was not serious, $CO_2$-NBW addition could alleviate the inhibition of VFAs, restore the pH to the normal range, improve the stability of the HSAD and can promote the consumption of soluble organics (Fig. 1 and Fig. 4). However, the addition of NBW could not change HSAD failure when severe inhibition occurs (for example, the content of VFAs was higher than 10,000 mg/L). In addition, after $CO_2$-NBW addition, the electric

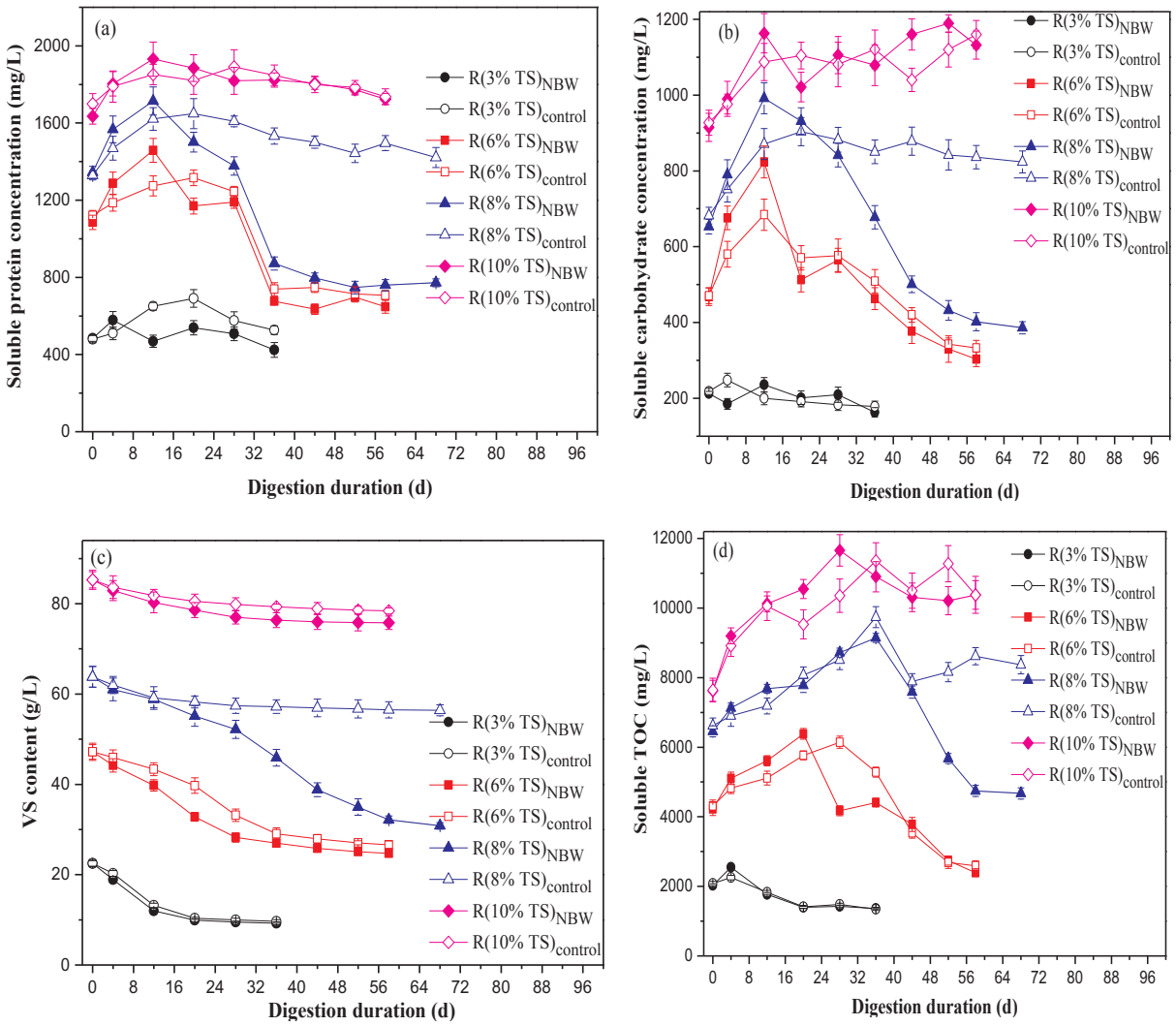



**Fig. 4.** Change of soluble protein concentration, soluble carbohydrate, soluble TOC.

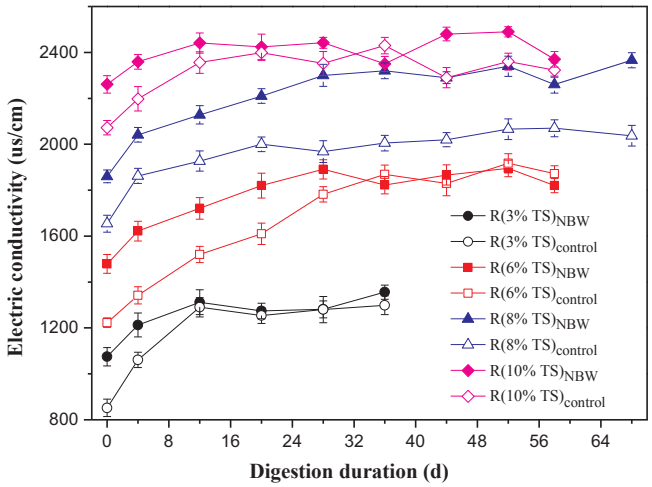



**Fig. 5.** Electric conductivity variation in all fermentation reactors during the AD process.

conductivity in reactors was increased (Fig. 5). The enhanced electric conductivity might imply the promoted mass transfer efficiency in the reactors of $R_{NBW}$. In addition, Yang et al. (2019) reported that NBW promoted enzymatic activity (protease, α-glucosidase, alkaline phosphatase and acid phosphatase) and hydrolysis of the organic solids by higher zeta potential and particle density. Methanogenic bacteria activity has also been proved to be improved by adding NBW in AD process (Wang et al., 2020).

Fig. 6 describes the possible mechanisms of applying NBW in the HSAD of SM. Unlike $R_{Control}$, the reactors of $R_{NBW}$ contains not only raw materials (PM and inoculum) and microorganisms but also large amount of NBW particles with higher zeta potential (negative), $T_2$ value and electric conductivity than $R_{Control}$. When PM is loaded into the reactors of $R_{NBW}$, NBW addition promotes the hydrolysis process of PM to soluble organics by increasing the activity of hydrolytic enzymes. In addition, NBW addition improves the mass transfer efficiency of nutrients from liquid to microbial cells and thus beneficial to quickly increase the activity of microorganisms. Meantime, NBW addition also improves the mobility of water molecules. The improved mass transfer efficiency of nutrients and the mobility of water molecules are conducive to the consumption of soluble organics by microorganisms, thereby accelerating the acidogenesis process and the production of intermediate products, such as VFAs. When the content of the intermediate products is lower than the threshold that NBW can withstand the inhibition, AD or HSAD succeeds. NBW addition improves the stability of AD or HSAD process. Moreover, since NBW addition can promote the activity of methanogens, methane production is also improved. However, when the content of the intermediate products is higher than the threshold, NBW addition is not enough to change the

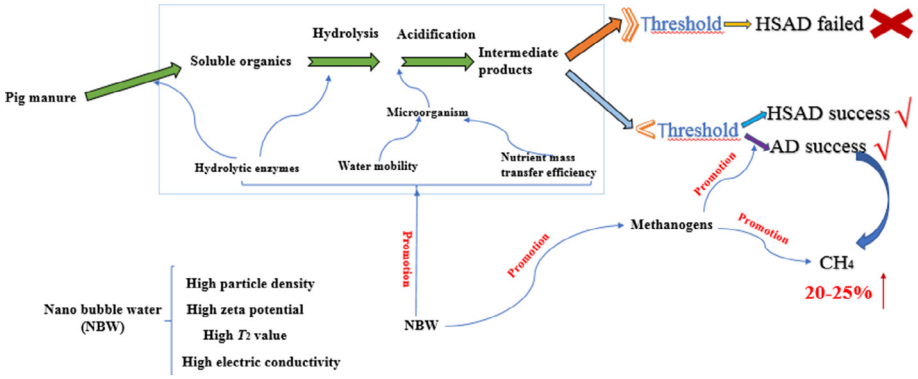



**Fig. 6.** The possible mechanisms for wet AD and HSAD under NBW addition.

failure of the HSAD. To reveal the whole mechanism of NBW effect on HSAD, further research is necessary, such as the inhibition threshold of VFAs and TAN and related gene expression under NBW addition.

## 4. Conclusions

To reveal the effect of NBW on HSAD of PM, this work evaluated the digestion stability and methanogenesis performance of HSAD under NBW addition. Results show that NBW addition improved the digestion stability, buffered pH and alleviated the inhibitions caused by the accumulation of intermediate products (such as VFAs). NBW addition also promoted the consumption of soluble proteins/carbohydrates. In addition, no matter in wet AD or HSAD process, methane production was enhanced under NBW supplementation. Therefore, NBW-based HSAD system is supposed to be capable of stable operation with increased methane production, which still deserves further research.

## CRediT authorship contribution statement

**Yujie Fan:** Investigation, Methodology, Writing - original draft, Data curation. **Zhongfang Lei:** Conceptualization, Writing - review & editing, Supervision. **Xiaojing Yang:** Formal analysis, Methodology. **Motoyoshi Kobayashi:** Resources. **Yasuhisa Adachi:** Resources. **Zhenya Zhang:** Conceptualization, Writing - review & editing, Supervision. **Kazuya Shimizu:** Conceptualization, Writing - review & editing, Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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